33kV OVERHEAD LINE FROM PARKSTON POWER STATION POLE FIM FA35 RECLOSER
2000-F12-SLD-EL-10001

F12-TR-012
33 / 11kV
5000kVA
Dyn11
z=7%
KNAN

F12-NR-012
PRIMARY CRUSHING 33/11kV TRANSFORMER F12-TR-012 NEUTRAL EARTHING RESISTOR
2000-EL011-B02-10004

30/0.05A EFCT
25A FOR 10s
254Ω +/-5%
SENSING
ER-15kV

240V SUPPLY
F12-DB-001 SWITCHROOM L&SP DB
2000-F12-TLD-EL-10101
F12DB001-R12
F12DB001-N12
HEATER

F12-SW-012
PRIMARY CRUSHING 11kV SWITCHBOARD PANEL A02
2000-EL003-F16-10005

EARTH BAR
QCE
BAD
PFV1
RD

BCT1 300 / 1A 2.5VA CL 0.2
BCT2 300 / 1A 2.5VA CL 0.2
BCT3 300 / 1A 2.5VA CL 0.2
BCT1 300 / 1A 2.5VA 5P20
BCT2 300 / 1A 2.5VA 5P20
BCT3 300 / 1A 2.5VA 5P20
SEL 735
SEL 751
2000-EL003-F16-10005 SHEET 16

-QAB 630A
RD
WH
BU
11kV 1250A 3-E 25kA FOR 3s

X4-21
X4-22
BCF2 (PART OF)
CONSTRUCTION NOTE 2

F12TR012-JB
PRIMARY CRUSHING TRANSFORMER F12-TR-012 PROTECTION DEVICES TERMINAL BOX
2000-EL011-D02-10001

PRV
PRESSURE RELIEF VENT TRIP ALARM
OIL LEVEL
OIL LEVEL ALARM (MINIMUM)
OIL LEVEL
OIL LEVEL ALARM (MAXIMUM)
OIL TEMP
OIL TEMPERATURE ALARM (100°C)
OIL TEMP
OIL TEMPERATURE TRIP (110°C)
WINDING
WINDING TEMPERATURE ALARM (110°C)
WINDING
WINDING TEMPERATURE TRIP (125°C)

F12-SW-012
PRIMARY CRUSHING 11kV SWITCHBOARD PANEL A02
2000-EL003-F16-10005

BCF1 (PART OF)
1535 1545 1595 1605 1625 1635 1685 1695
X2 10 55 14 59 15 60 16 61
48VDC CONTROL
D10 IN409 TRANSFORMER PROTECTION TRIP
D14 IN413 TRANSFORMER OIL LEVEL MAXIMUM ALARM
D15 IN414 TRANSFORMER OIL TEMPERATURE ALARM
A10 IN101 TRANSFORMER WINDING TEMPERATURE ALARM
SHEET 13

F12SR001-LCS
SWITCHROOM EMERGENCY SHUTDOWN PUSHBUTTON RELAY K2 (LOCATED IN AIR LOCK)
2000-F12-TER-EL-10202

1760 1780 1800
X3 23 33
X5 32 34 50 51 63 64 30 31
SWITCHROOM EMERGENCY SHUTDOWN PUSHBUTTON CONNECTION TO THE VCB 1st TRIP CIRCUIT SHEET 8
E-SHUT NOTE 5
F12SR001LCS-48VDC+
F12SR001LCS-48VDC-
BCF1 (PART OF) C03 OUT302 C04
PROTECTION TRIP UPSTREAM SHEET 15
UPSTREAM SHUTDOWN MARSHALLIN TERMINALS

F12-BC-001
11kV SWITCHROOM 48VDC BATTERY CHARGER
2000-F12-TLD-EL-10001

POLE FA35 RECLOSER (NOTE 4)
K1 RECLOSER INTERPOSING TRIP RELAY
1915 1935
INPUT 1 DOWNSTREAM SWITCHROOM INTERTRIP
CONSTRUCTION NOTE 1
INPUT 2
INPUT 3
I/O MODULE 2000-CN008-D02-30001

F12-DCS-401
415V SWITCHROOM DCS PANEL
2000-F12-TER-IC-10014
F12SW012OP+48VDC
F12SW012OP
F12SW012CL+48VDC
F12SW012CL
01/03/01
01/03/02
DCS REMOTE CB OPEN
DCS REMOTE CB CLOSE

NOTES

1. WIRE NUMBER FERRULES ON TERMINAL STRIP SHALL BE PREFIXED WITH EQUIPMENT NUMBER e.g. F12SW012-1200, UNLESS OTHERWISE NOTED.
2. EACH WIRE SHALL BE IDENTIFIED AT BOTH ENDS WITH FERRULES.
3. ALL ALARMS AND TRIPS SHALL BE FAIL-SAFE. NOTE THAT THE PROTECTION RELAY INPUTS SHALL BE INVERTED, I.E. ALARM AND TRIP WHEN CONTACT IS OPENED.
4. NOTE THAT CROSS SECTIONAL AREA OF CABLES TO THE OVERHEAD LINE RECLOSER ARE INCREASED IN SIZE - REFER TO THE CABLE SCHEDULE.
5. E-SHUT CIRCUIT IS FAIL SAFE. NORMALLY CLOSED CONTACTS WILL BE OPEN WHEN THE E-SHUT BUTTON IS UN-ACTIVATED AND THE CIRCUIT HEALTHY.

CONSTRUCTION NOTES

1. INSTALL NEW RELAY K1 AND UTILISE SPARE X5 TERMINALS FOR TRIPPING UPSTREAM 33kV OHL RECLOSER.

   INSTALL NEW RELAY K1 INTO POLE FA35 RECLOSER CONTROL PANEL.

   CONFIRM TERMINALS X5:50 & X5:51 ARE SPARE.

   TERMINATE NEW 2C+E CABLE IN K1:A1 AND K1:A2 TERMINALS.

   INSTALL WIRING FROM K1 N/O CONTACT TERMINAL 33 TO INPUT 1 OF I/O MODULE AND WIRING FROM K1 N/O CONTACT TERMINAL 34 TO COM OF I/O MODULE.

   REFER TO ABB DRAWING 2000-EL003-D03-10001 SHEET 4 FOR X5 TERMINALS.
2. CONFIRM TERMINAL X4-23 IS SPARE.
   INSTALL WIRING FROM TERMINAL X4-23 TO (SE330 AU) BCF2 TERMINAL R:6.

LEGEND

- xx/xx/xx — DCS DIGITAL INPUT
- xx/xx/xx — DCS DIGITAL OUTPUT
- xx/xx/xx — DCS ANALOGUE INPUT
- xx/xx/xx — DCS ANALOGUE OUTPUT
- xx/xx/xx — DCS ANALOGUE RTD INPUT
- ⊗ — 11kV SWITCHBOARD TERMINAL 1
- VSD TERMINAL
- FIELD TERMINAL
- DCS OR I/O TERMINAL
- MARSHALLING TERMINAL
- (x) — DEVICE TERMINAL

PANEL A02

DRG No.	REFERENCES
2000-EL003-F16-10005	F12-SW-012 ABB SCHEMATIC DIAGRAM
2000-F12-TER-EL-10103	F12-SW-012 11kV SWITCHBOARD - PANEL A02 (INCOMER) TERMINATION DIAGRAM SHEET 1
2000-F12-SLD-EL-10001	F12-SW-012 11kV SWITCHBOARD SINGLE LINE DIAGRAM
2000-F00-STD-EL-10001	GENERAL NOTES AND STANDARD DRAWINGS INDEX

REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
1	18JUL25	FAT REDLINES INCORPORATED				MB	
0	10MAR25	APPROVED FOR CONSTRUCTION					

NORTHERN STAR RESOURCES LIMITED

PRIMERO
primero.com.au | DESIGN | CONSTRUCT | OPERATE
PRIMERO PROJECT NUMBER: 23517

INFORMATION AND REPRESENTATIONS ON THIS DRAWING ARE THE INTELLECTUAL PROPERTY OF PRIMERO UNLESS SPECIFICALLY IDENTIFIED OTHERWISE. ALL RIGHTS INCLUDING USE RESERVED

PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: PRIMARY CRUSHING - F12-SW-012
F12-SW-012 INCOMER
SCHEMATIC DIAGRAM

SHEET	SCALE	DRG No.	REVISION
A1	NONE	2000-F12-SCM-EL-10103	1

O:\CAD PROJECTS\23517 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\ELV\F12 - PRIMARY CRUSHING\2000-F12-SCM-EL-10103.DWG 13AUG25 11:13AM HOTSEAT04